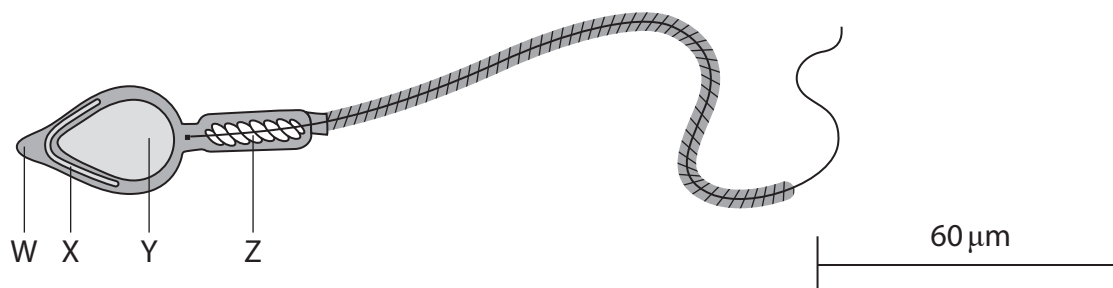


7 A sperm cell is a specialised cell.

(a) The diagram shows a sperm cell.



(i) How many of the labelled structures contain DNA?

(1)

- ☐ A 1
- ☐ B 2
- ☐ C 3
- ☐ D 4

(ii) Sperm cells contain mitochondria.

Describe the function of mitochondria in the movement of sperm cells.

(2)

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(iii) Calculate the magnification of the sperm cell shown in the diagram.

(2)

Answer



P 6 1 4 7 0 A 0 2 3 3 2

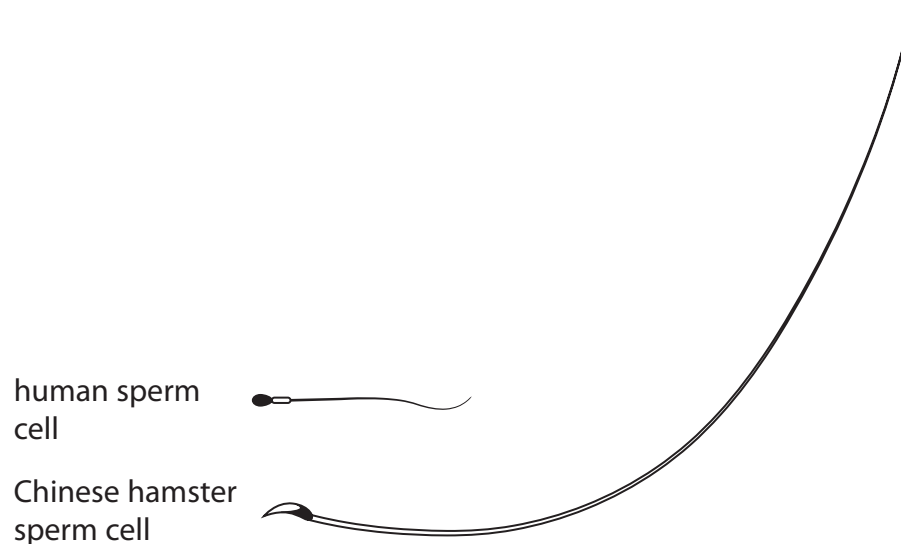
(b) The photograph shows a Chinese hamster.



Magnification $\times 1$

A female Chinese hamster mates with many males in a short period of time.

The diagram shows the relative size of a Chinese hamster sperm cell compared with a human sperm cell.



Suggest why the Chinese hamster sperm cell has such a long flagellum.

(3)

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(c) Explain the role of the cortical reaction in the process of fertilisation in mammals.

(3)

(d) After fertilisation, the egg cell divides by mitosis to form a blastocyst.

During this process, totipotent cells become pluripotent.

Describe how a totipotent stem cell becomes a pluripotent stem cell.

(3)

(Total for Question 7 = 14 marks)



P 6 1 4 7 0 A 0 2 5 3 2